

The surreal and surcomplex reports

Twenty-six independent mathematical research packages,
on the surreal field, its algebraic closure, its quaternion algebra,
their reach into physics, and the foundations and computation of all of them

September 2026

Scope

This document catalogues the twenty-six research packages stored under `docs`. Each entry names the problem the report attacks, the result it claims, and the delivery it arrived in.

The collection divides in five, and so does the directory tree. Eight reports, under `docs/surreal`, are about the surreal field **No** itself: birthdays and their behaviour under multiplication, the option graphs of surreal forms, genetic functions and gaps, two questions of Lipparini about evaluating formal sums, the rigidity of exponential automorphisms, and Gamma functions. Thirteen, under `docs/surcomplex`, are about the algebraic closure $\mathbf{No}[i]$: the theory of holomorphic functions over it, its differential algebra, its rank-one analytic geometry, its finite matrices, its local and quasi-periodic dynamics, its nonabelian support obstructions, and its entire functions at arbitrary valuation rank. One, under `docs/surquaternions`, leaves commutativity behind for Hamilton's division algebra $\mathbb{H}_{\mathbf{No}}$ over **No**. One, under `docs/physics`, is the only report here whose subject is not mathematics: it asks what surreal scalars do and do not buy in general relativity, and answers negatively. The last three, under `docs/foundations-and-computation`, stand outside all of these: they are about the subject rather than inside it, asking in what foundation this mathematics legally lives, how one computes with it exactly, and which surreal numbers are computable at all.

These reports were previously part of a larger collection, in a repository whose subject was elsewhere; they were moved here with their git history. That is why an entry may refer to a merge, a duplicate sweep or a delivery number: those describe how the report reached its present form, and the record is kept rather than rewritten.

Each directory holds a typeset article with its L^AT_EX source, a `README.md`, and in most cases verification code together with the recorded output of running it. The source manuscripts a merged report was assembled from are not distributed here. What is kept is the provenance record: each merged report states how many manuscripts it was merged from, what each contributed, and where the merge had to choose between them, and the disposition of every source result is recorded in that material. These are AI-assisted research drafts. Some claims and prerequisites have checked Lean proofs, mapped precisely in `FORMALIZATION.md`; no complete verification of the collection is claimed. This catalogue describes the reports rather than certifying them. The cross-report `NOTATION.md` explains shared names and distinctions between local conventions.

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1 The surreal field

Eight reports about **No**. Two questions of Lipparini on evaluating formal sums; two independent halves of Gonshor's product-birthday conjecture, on domains neither of which contains the other; the subgraph question from Roughan's paper on birthday arithmetic; a genetic counterexample separating pointwise derivatives from differences of constants; a rigidity theorem for exponential automorphisms; and a sharp convexity classification of surreal Gamma functions. The first and

the fourth each arrived as a pair of archives and were merged; the last two are standalone and independent of everything else here.

1. A square-root obstruction to evaluating increasing Hahn series at ω

[surreal/hahn-evaluation-at-omega/article.pdf](#)

from [hahn_evaluation_research.zip](#) + [reversed_hahn_series.zip](#)

Answers Problem 7.7 (p. 38) of Lipparini (arXiv:2505.00424v2) negatively: there is no ring homomorphism from the field of set-supported real-coefficient Hahn series with strictly increasing surreal exponents into the surreals sending x^a to ω^a . The certificate is $B \in \mathbb{Q}[[x]]$ with $B^2 = 1 - x$, built by the recursion $b_n = -\frac{1}{2} \sum_{k=1}^{n-1} b_k b_{n-k}$: any such map would force a square to equal $1 - \omega < 0$. The homomorphism is applied only to the finite identity, so no continuity, valuation, order preservation or preservation of infinite sums is assumed. Adds the exact classification of admissible images of the variable as the infinitesimals, the monomial classification over an arbitrary ordered abelian group, and order-uniqueness on the source field. *Merged; the core certificate was identical in both originals and is proved once, but the two toolkits are kept — in particular a direct nonnegative-coefficient semiring certificate. Restriction and extension by differences relate the ring and semiring maps; the two witnesses give distinct elementary proofs.*

2. Gonshor's Product-Birthday Conjecture for ordinal-support series

[surreal/gonshor-product-birthdays/surreal_product_birthdays.pdf](#)

from [gonshor_product_birthdays_research.zip](#)

Proves Gonshor's bound $b(xy) \leq b(x) \otimes b(y)$ on the ring of Conway normal forms $\sum_{\alpha \in S} c_\alpha \omega^{-\alpha}$ with S an arbitrary set of ordinals, so transfinite and uncountable supports are allowed. Supplies an exact birthday formula by transfinite induction over the support with maintained prefix bounds and an explicit limit stage, and a complete classification of the equality cases. The route is a leading-Cantor-degree gap, which also gives that the degree of the birthday never rises under multiplication. The conjecture for arbitrary surreal numbers is not solved. The domain is incomparable with that of the Laurent-series report [surreal/gonshor-laurent-birthdays](#): it excludes ω and every positive power.

3. A Laurent-series case of Gonshor's birthday conjecture

[surreal/gonshor-laurent-birthdays/article.pdf](#)

from [Gonshor_Laurent_Birthday_Research.zip](#)

Proves the same bound $b(xy) \leq b(x) \otimes b(y)$ on the canonically embedded field $\mathbb{R}((\omega^{-1}))$, which unlike the ring of [surreal/gonshor-product-birthdays](#) contains ω and its positive powers but restricts supports to order type at most ω on the integer lattice. The route is a bilinear budget: an exact splitting $b(x) = b(P) \oplus b(u)$ into polynomial and infinitesimal parts, three separate product estimates for the resulting cross terms, and a finite-jet stabilisation carrying the finite bound to infinite series without any continuity claim. Also treats finite normal forms with arbitrary ordinal exponents, derives exact birthdays for finite and infinite Laurent series, and proves a reciprocal corollary. Neither this theorem nor that one contains the other; on their common corner $\mathbb{R}[[\omega^{-1}]]$ the two agree and are proved twice.

4. Canonical forms need not be subgraphs

[surreal/canonical-forms-need-not-be-subgraphs/surreal_graphs.pdf](#)

from [surreal_graphs_research.zip](#) + [surreal_sparse_graphs.zip](#)

Answers negatively the question in the Figure 3 caption of Roughan, *Surreal Birthdays and Their Arithmetic* (arXiv:1810.10373v2, p. 7): the canonical form of a surreal value need not be a subgraph of every other form of that value. Five vertices and rank three are both least possible, and there are exactly sixteen five-vertex counterexamples, eight triangle-free, four of value $1/2$. Adds cycle representations of $1/2$ at every admissible length, the exact extremal function, a planar construction of maximum degree three and arbitrarily large girth, classifications of which canonical graphs are unavoidable, and transfinite witnesses. *Merged; the two originals exhibited genuinely different five-vertex witnesses — one passing through a vertex of value -1 , one with no*

negative value at all — and both are printed in full, as are two odd-cycle families, two high-girth constructions and two proofs of the Fibonacci bound.

5. A countable genetic cut across the finite–infinite gap

[surreal/genetic-gaps-and-primitives/article.pdf](#)

from [genetic_gap_counterexamples.zip](#)

With $q(t) = t/(1+t^2)$ and $B(x) = 1 - \{q(n-x) : n \in \mathbb{N} \mid \}$ as a Conway cut, proves that B is the indicator of the positive infinite surreals, is genetic under the set-sized option constructors of the cited sources, and has pointwise derivative zero everywhere. Consequently x and $x - B(x)$ are genetic primitives of 1 agreeing at every real input that do not differ by a constant, the latter also an increasing bijection. The sublevel class $\{x : 0 < x < \omega, B(x) \leq \frac{1}{2}\}$ is nonempty and bounded yet has no surreal supremum, so neither end gap supplies one.

6. A Number-Valued Broadcast Sum of Surreal Sequences

[surreal/broadcast-sum-of-surreal-sequences/article.pdf](#)

from [surreal_broadcast_sum.zip](#)

Analyses the sign-truncation game of Lipparini (arXiv:2505.00424v2, Remark 7.6(3), in connection with Problem 7.5). Proves the game always has a surreal-number value, that it agrees with finite surreal sums and with Lipparini’s ordinal operation on their respective domains, and that it fails weak coordinatewise monotonicity, with an explicit family $s[n] = 3 \cdot 2^{-n-d-1}$ on a subset E and 2^{-n-d} off it realising the failure. The underlying rule is Lipparini’s; the name and the notation are local to the report.

7. Cofinal displacement and the rigidity of exponential automorphisms

[surreal/exponential-automorphism-rigidity/article.pdf](#)

from one manuscript, delivered 21 September 2026

Let F be an ordered field with a total increasing ordered exponential $E : (F, +) \rightarrow (F_{>0}, \cdot)$ that is *onto*, and w a nontrivial convex valuation — no Hahn workspace, no completeness, no saturation, no set-size hypothesis. The engine is one finite lemma: for a unital field endomorphism σ the displacement $D(x) = \sigma(x) - x$ is additive but is *not* a derivation, the correct rule being the twisted identity $D(ab) = \sigma(a)D(b) + bD(a)$, with $\sigma(a)$ and not a ; from $D(a) = d \neq 0$ the single probe pair $b = 2B(1 + |\sigma(a)|)/|d|$ and ab forces $\max\{|D(b)|, |D(ab)|\} > B$. Two probes, no transfinite construction, and neither surjectivity nor order preservation of σ is needed, so self-embeddings and class-sized fields are covered. Consequences: a nonidentity exponential automorphism stabilizing $\mathcal{O}_w$ displaces value-group elements cofinally *and* coininitially, so $\rho_w : \text{Aut}_{\text{exp}}(F; \mathcal{O}_w) \rightarrow \text{Aut}(\Gamma, +, <)$ is injective; coarsening rigidity on Γ/Δ , with stabilization of Δ an explicit hypothesis; and on $K = F(i)$ the intrinsic logarithmic modulus $L(z) = \log \sqrt{z\bar{z}}$ — real logarithm only, no branch, no argument — gives $\text{Aut}_L(K) \cong \text{Aut}_{\text{exp}}(F) \times C_2$. The bridge is $\nu = w \circ E$, additive, surjective and order-reversing, whose kernel is a proper convex additive subgroup; such a subgroup of an ordered field is bounded by a *field* element and not by an integer or a real, which is the whole point — the finite surreals are bounded by ω , and the contradiction is the probe lemma run at $B = \omega$. Specialized to **No**: every exponential 1-automorphism is the identity, offered as a negative answer to Question 5.4 of Kaplan–Krapp–Serra, *Decomposing the automorphism group of the surreal numbers*, arXiv:2509.22374v3. *The version pinning is deliberate and should stay prominent*: the claim is against one arXiv version, the reading of the question has not been checked anywhere else in this collection, and Gonshor’s exponential and the natural valuation are cited rather than reproved. Section 6 separates ordered-Hahn symmetry from exponential symmetry by two disjoint mechanisms, every one of which does lift as an ordered field automorphism — so cofinal displacement is necessary and provably *not* sufficient for exponential lifting. This is a third thing called “rigidity” here: not the all-scale polynomial rigidity of [surcomplex/analysis](#), not the scalar rigidity of [surreal/gamma-functions](#), but faithfulness of ρ_w .

8. Gamma functions over the surreals: a sharp convexity classification

[surreal/gamma-functions/article.pdf](#)

from one manuscript, delivered 21 September 2026

Write $M(x)$ for the leading Conway monomial of a positive infinite surreal with its real coefficient stripped and $\text{pp}(x)$ for the purely infinite part; for an arbitrary class assignment a on positive infinite monomials, no regularity assumed, set $h_a(x) = a(M(x)) \text{pp}(x)$ and $L_a = L_0 + h_a$ over the Taylor–Stirling log-Gamma baseline. The main theorem is a two-sided sharp threshold: L_a is convex on all of $\mathbf{No}_{>0}$ iff $ma(m)$ is finite for every positive infinite monomial m , and in that case it is *strictly* convex — pointwise in m , no uniform real bound across monomials, both signs allowed, and no merely-convex member exists. Every member of the family, convex or not, preserves $\Gamma_a(1) = 1$, the recurrence $\Gamma_a(x+1) = x\Gamma_a(x)$, every finite Gauss multiplication formula, every normalized finite-translation germ and all positive-order fine derivatives of log-Gamma, so the literal surreal Bohr–Møllerup conditions do not give uniqueness: $a_\lambda(m) = \lambda e^{-m}$, $\lambda \in \mathbb{R}$, gives pairwise distinct strictly log-convex extensions whose logarithmic and relative differences are smaller than *every* ordinary inverse power at *every* infinite argument, the log-difference at ω being exactly $\lambda \omega e^{-\omega}$, and adding all ordinary-order Stirling envelopes does not restore uniqueness. Absolute Gamma differences need not be small: at ω the difference $\Gamma_1 - \Gamma_0$ exceeds every ordinary power of ω . The convexity proof uses no second derivatives and no mean value theorem: it works with the tangent gap $D_F(y, x) = F(y) - F(x) - F'(x)(y - x)$ through four exhaustive relative-scale regimes, the margin $(y - x)^2/(2x)$ being exactly what generates the $O(m^{-1})$ threshold, and the mechanism is then isolated as a reusable scale-margin gauge principle. A positive second derivative does *not* imply convexity in this calculus — the gauge with value one on one monomial and zero elsewhere has all fine derivatives equal to the baseline’s and a negative tangent gap — because such gauges are finely locally constant and convexity cannot be decided locally. Compatibility with the Berarducci–Mantova derivation forces h_a into $\ker \partial = \mathbb{R}$ and kills every member but $a = 0$, yet a different real-valued non-flat gauge survives, so *scalar compatibility alone is not a uniqueness characterization*. Closes with an exact finite-phase domain on the tube $\text{Re}(z) > 0$, $\text{Im}(z)$ finite, using the finite-angle exponential of the imaginary log-Gamma value, distinct from choosing a primitive in [surcomplex/differential-equations](#). *Not offered as a resolution of a named published problem*, and the classification is exact only for the named monomialwise family. Its own audit read an earlier state of this catalogue; its eighteen-package count is a historical description, not the current inventory.

2 The surcomplex plane

Thirteen reports about $\mathbf{No}[i] = \mathbf{No}[i]$. Seven are about holomorphic function theory, rest on one framework, and are all merges: twenty-seven manuscripts reduced to seven documents. The other six came later and are listed after them — four merges, of seven manuscripts on differential algebra, seven on dynamics and normal forms, two on nonabelian support obstructions and two on entire functions at arbitrary rank, together with two standalone reports, on rank-one analytic geometry and on finite matrices. Each of the six was written against this repository and each names the gap it fills.

*Read **analysis** first.* It is the foundation the other six continue. In the *full fine topology*, every set-sized subset is closed and discrete and every convergent set-indexed net is eventually constant. Universe-relative formulations require smaller-universe subsets and indices. These conclusions do not apply to the intrinsic valuation topology of a fixed Hahn field: $t^n \rightarrow 0$ in $\mathbb{C}((t^{\mathbb{Q}}))$. Strong Hahn summability is a separate support condition; even in rank one it need not give convergence of finite subsums, as the foundations report’s $\sum t^{n/(n+1)}$ demonstrates. Contour integration in the analytic reports is defined coefficientwise on ordinary complex paths.

*Then read the ring table in **analytic-geometry**, before anything else.* Four coefficient rings circulate through this material under nearly identical notation — the radius-free $\mathbb{C}\{z\}((t^{\Gamma}))$, the

fixed-polydisk $\mathcal{O}(D)((t^\Gamma))$, the common-domain germ ring, and the formal $\mathbb{C}[[z]]((t^\Gamma))$. These are not an interchangeable chain. For $\eta > 0$, the family $F(z) = \sum_{r \geq 1} t^{r\eta}/(1 - rz)$ lies in the radius-free germ ring but not the common-domain germ ring: its coefficient poles $1/r$ approach zero. This witnesses that particular strict inclusion. The report gives separate fixed-domain obstructions and explains which local proofs require shrinking. The remaining five reports can be read in any order.

9. Surcomplex analysis: holomorphic functions on $\mathbf{No}[i]$

[surcomplex/analysis/article.pdf](#)

from `surcomplex_analysis.zip` and eight companion archives

One article assembled from nine manuscripts, all delivered under the same name on 21 September 2026 as independent attempts at one subject: a theory of holomorphic functions over $\mathbf{K} = \mathbf{No}[i]$ built on Hahn-supported series. On the convergence-free base described above the article develops germs, the coherent algebra $\mathcal{H}(U)$ over an ordinary domain, Weierstrass preparation and infinitesimal zero clusters, a residue calculus, the global principles, exponentials and logarithms, and an all-scale rigidity theorem. **Nine independent runs at a foundation do not agree the way nine runs at a theorem would, and the merge turned on that.** Of seventeen load-bearing notions, six are genuinely inequivalent across the sources, and a theorem proved under one is false under another: there are three inequivalent function classes (the identity theorem, the global maximum principle, Liouville and the open mapping theorem hold for coherent sections and fail for germs; sharp Schwarz and Schwarz–Pick hold for lifts and fail for coherent sections), and three inequivalent residues, two of which disagree numerically — for $1/(\zeta - \varepsilon)$ with ε infinitesimal the cluster residue at the ordinary centre 0 is 1 while the local residue at 0 is 0, the actual pole sitting at ε . A single ordinary bound in Liouville gives a characterization and not constancy: tz is real-bounded by 1 on the whole finite plane and is not constant. Eleven further conflicts are resolved and recorded, including two functions both called “the” logarithm that differ by exactly $2\pi i$. All nine verification programs are kept, and the disposition of every source result is recorded; the deduplication is in the article. Its `MERGE_NOTES.md` is worth reading before the article.

10. Infinitesimal analytic geometry over the surcomplex numbers

[surcomplex/analytic-geometry/article.pdf](#)

from `Surcomplex_Analytic_Geometry.zip` + `surcomplex_finite_geometry.zip`

The ring theory of the infinitesimal neighbourhood — the monad $\mathfrak{m}_K^n$ of an ordinary point — rather than the deformation theory of the reports that follow. Proves, *separately of each of* the radius-free $A_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$ and the common-domain $\mathcal{R}_n$, that it is a Noetherian Jacobson domain of dimension n whose maximal ideals are exactly the infinitesimal evaluation ideals $\mathfrak{m}_a = (z_1 - a_1, \dots, z_n - a_n)$ with residue field K , regular local at each, with completion $K[[X_1, \dots, X_n]]$ — and that neither is local, each recording every infinitesimal point at once. Carries Weierstrass division and preparation, a weak and a strong Nullstellensatz, Noether normalization, finite projection, contour residue duality, Hensel–Rouché, and workspace invariance: every common zero over any larger divisible workspace already lies in K_Γ^n with unchanged multiplicity, so the count accounts for all zeros in the whole proper class $\mathbf{No}[i]$. The nonzero-value-group hypothesis is essential and not aesthetic — for $\Gamma = \{0\}$ the strong Nullstellensatz fails at $I = (0)$. *This is the report that fixes the four-ring vocabulary and proves the inclusions strict; statements holding over both A_n and $\mathcal{R}_n$ are stated twice with both proofs, since the two proofs use different resources and neither transfers.*

11. Finite Hahn deformations: division, conservation of multiplicity, residue duality

[surcomplex/finite-deformations/article.pdf](#)

from `surcomplex_finite_deformations.zip`, `_finite_fibres`, `_finite_maps`, `_intersections`, `_research`

Five manuscripts on one theorem: an ordinary holomorphic complete intersection f on a fixed polydisk, with the origin as its only common zero and colength μ , keeps its entire finite geometry under any Hahn perturbation $F = f + E$ of well-ordered *strictly positive* error support — on the unshrunk domain, with an explicit support certificate $\text{supp} \subseteq \text{supp}(G) + M$, M the monoid generated by the error support. The remainder of division is unique; the witnesses need not be,

their ambiguity being a Koszul syzygy carried by the same certificate. Gives the integral structure $H^+_{\Gamma}(D)/(F) \cong (K^\circ)^\mu$ over a valuation ring that may have high-rank value group and need not be Noetherian, with nothing completed; conservation of multiplicity at the *actual* displaced surcomplex zeros rather than their standard parts, $\sum_{\zeta} \mu_{\zeta} = \mu$; and a coefficientwise residue Res_F — defined by a series, not by a surcomplex contour integral — whose Gram determinant is a unit even at a collision and which satisfies $\text{Res}_F(G \cdot J_F) = \text{Tr}(M_G)$. The separate residue formula, when all zeros are simple, is $\text{Res}_F(G) = \sum_{\zeta} G(\zeta)/J_F(\zeta)$; for nonnegative-support G , individual terms can be infinite while the total is finite. Also Hartogs, characters and local algebras, spectral invariants, trace-Jacobian and Bézoutian, monodromy and stability. For $\mu = 1$, the inverse-map proof constructs one uniform Hahn support before evaluation. The engine throughout is positive-support Neumann inversion; ordinary analytic convergence occurs in finite-parameter comparisons, while the Hahn constructions use strong support summation.

12. Jordan separation, Cauchy theory, and Stokes calculus on the surcomplex plane

[surcomplex/contours-and-stokes/article.pdf](#)
from `surcomplex_contours.zip` and two same-named archives

Three manuscripts whose titles differ by one word — Stokes *Calculus, Integration, Formulae* — over one body of work. All three refuse to transfer the Jordan–Cauchy–Stokes package to $\mathbf{No}[i]$ as a unit, and all three split it the same way: a coarse Jordan separation theorem in the standard-part topology whose separator is the whole boundary monad rather than a curve; an imported semialgebraic Jordan theorem; a Hahn-coefficient de Rham complex with coefficientwise Stokes, extended to infinitesimally deformed chains; and the holomorphic consequences — Cauchy, endpoint displacement, integer winding, and microscopic circle and torus realizations of the perturbation residue — with an algebraic rational contour pairing characterized uniquely by vanishing on exact differentials and normalizing $dz/(z - a)$ to $2\pi i \cdot \text{wind}_{\text{sa}}$. Adds a purely algebraic third chain category over any characteristic-zero field, a support-preserving Poincaré lemma, an order-valued ML estimate, and a root-of-unity/Schirelman section. This is where the question the deformation report leaves open is answered: supplying an actual surcomplex contour that represents the residue series. *Two thirds of each manuscript was the same mathematics in the same order; 82 numbered results became 59, by duplicate elimination only.*

13. Global support obstructions: moving divisors, Mittag–Leffler, and a Picard dichotomy

[surcomplex/global-divisors/article.pdf](#)
from `surcomplex_global_support`, `_global_theorems`, `_compact_curves`

The local-to-global theory: when do infinitely many individually legitimate infinitesimal prescriptions admit one coherent global realization over the ordinary complex plane? The sharp answer for divisors is a *symmetric-support* criterion — a cluster family $P_c(u) = u^{m_c} + \sum_{j < m_c} a_{c,j} u^j$ over a closed discrete ordinary $A \subset \mathbb{C}$ is the exact zero divisor of a nonzero global section iff $\bigcup_c \bigcup_j \text{supp}(a_{c,j})$ is well ordered: the supports of the symmetric coefficients, not of separately chosen roots. Hence the two discriminating examples — $n + t^{1/n}$ alone is not realizable, its descending symmetric support $\{1/n\}$ failing, while *all* roots of $(z - n)^n = t$ are, symmetric support the singleton $\{1\}$ — and hence a globally principal effective divisor with a nonprincipal effective sub-divisor. Also Mittag–Leffler at fixed centres and for moving poles, a support-restricted Chinese remainder theory with an explicit Neumann right inverse, an exact Cousin criterion in terms of singular support, and the Picard dichotomy: every ordinary-base stalk is a principal ideal domain, nonlocal when $\Gamma \neq 0$; $\text{Pic}(\mathbb{C}, \mathcal{H}_{\Gamma}) = 0$ iff $\Gamma = \{0\}$ or $\Gamma = \mathbb{Z}\delta$, and otherwise $\dim_{\mathbb{C}} \text{Pic} \geq 2^{\aleph_0}$, with equality for countable noncyclic Γ . A third manuscript studies a compact connected ordinary Riemann surface X of genus g . Its Picard group splits into $H^1(X, \underline{\Gamma}) \oplus \text{Pic}(X) \oplus (H^1(X, \mathcal{O}_X) \otimes_{\mathbb{C}} \mathbf{m}_{\Gamma})$. The valuation class is exactly the obstruction to a nonzero coherent meromorphic section, even in arbitrarily high degree. In the zero-valuation sector a finite matrix with positive-support entries computes cohomology and gives Riemann–Roch. A root-free Newton-sum Abel functional decides principality of finite signed clusters. The one-scale comparison has kernel $H^1(X, \mathcal{O}_X) \otimes_{\mathbb{C}} \bigcap_n t^{n\delta} \mathcal{O}_{\Gamma}$;

it is injective exactly when $g = 0$ or δ is cofinal in Γ . Any prescribed set of surreal cutoffs can be defeated after enlarging the workspace; all positive cutoffs detect classes in a fixed nonzero workspace. Coordinate charts transfer the local PID argument to X , but the global plane and compact classification theorems retain their different base hypotheses.

14. Surcomplex polynomial algebra: root geometry, Newton profiles, residue duality

[surcomplex/polynomial-algebra/article.pdf](#)

from `surcomplex_polynomial_algebra.zip` and two same-named archives

Finite-degree polynomial algebra over $\mathbf{No}[i]$, worked inside set-sized algebraically closed Hahn workspaces and keeping two geometries deliberately apart throughout: the surreal-valued modulus, whose disks are ordered, and the natural Hahn valuation, whose balls have arbitrary ordered-group rank. A Rouché theorem in each. Covers the finite polynomial algebra and the fundamental theorem there, CRT/Hermite jets, resultants and discriminants, ordered root geometry (root bounds, Gauss–Lucas, Jensen disks, a sharp Schoenberg second-moment inequality with its equality case), real-closed-field transfer with Hermite signature counting, exact root loci under coefficient uncertainty in both geometries, Newton profiles over an arbitrary ordered exponent group, image balls and branch values, exact Rolle counts and the finite root-cluster tree, support-controlled coprime Hensel factorization with a fixed-domain parameter form that shrinks no neighbourhood, and universal residue duality with the trace identity. *Three perturbation theorems with three different hypotheses are kept separate rather than collapsed, as are two discriminant layer formulas and two mapping theorems, with their agreement proved; the alternative was to lose what each source reached that the others did not.*

15. Trigonometry on the surcomplex plane

[surcomplex/trigonometry/article.pdf](#)

from `surcomplex_trigonometry.zip` and two same-named archives

Its organizing theorem is that every surcomplex direction has a *finite* surreal angle: $\text{cis} : (\mathcal{O}_{\mathbb{R}}, +) \rightarrow (T(\mathbf{No}), \cdot)$ is surjective with kernel exactly $2\pi\mathbb{Z}$, so $\mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z} \cong T(\mathbf{No}) \cong T(\mathbb{R}) \times (\mathfrak{m}_{\mathbb{R}}, +)$, and the whole of plane, spherical and hyperbolic triangle geometry is available at arbitrary surreal length scale without ever choosing a value for $\sin \omega$. From it: canonical sine and cosine by Hahn summation, the unit circle with its branch conventions and projective half-angle chart, inverse functions on their full surreal domains, the angle-sum, cosine, sine, tangent, half-angle, Heron, incircle, Euler, angle-bisector, cevian, Ceva, inscribed-angle and Ptolemy theorems, an exact valuation dictionary between sides, angles and area with a quadratic triangle-inequality defect formula and two flat-triangle theorems, a rank-two collision-stable intersection algebra, the canonical complex extension to the strip where sine is surjective with completely determined fibres, finite Fourier inversion, Parseval and a surcomplex Fejér–Riesz factorization, and the classification of all global phase extensions, with the Ehrlich–Kaplan normalization placed inside that classification. Two coherence obstructions close it: there is no coherent bounded sine of infinite frequency, and all-scale coherence forces a polynomial. *No mathematical disagreement was found between the three sources; the duplication was near-total on the first two thirds and the divergence near-total on the last.*

16. Differential algebra and differential equations over the surreal and surcomplex numbers

[surcomplex/differential-equations/article.pdf](#)

from `surreal_differential_algebra.zip` and six companion archives

Seven manuscripts on the Berarducci–Mantova derivation ∂ on $\mathbf{No}$, normalized by $\partial\omega = 1$, and its unique extension to $\mathbf{No}[i]$ — unique because differentiating $i^2 + 1 = 0$ forces $\partial i = 0$. The organizing identity is exact: the image of the logarithmic derivative is $\mathbf{No} + i\mathcal{A}$ with $\mathcal{A} = \partial\mathfrak{m} = \partial\mathcal{O}$, so $\partial y = (a + ib)y$ has a nonzero surcomplex solution exactly when b has a *finite* primitive. Two things carry the subject. The two printed forms of that image agree only through the proved lemma $\partial\mathcal{O} = \partial\mathfrak{m}$, never as notational variants. And $\mathcal{A}$ is a *strict* subclass of $\mathfrak{m}$: ω^{-1} is infinitesimal while its primitive $\log \omega$ is infinite, so $\partial y = i\omega^{-1}y$ has only the zero solution — collapsing $\partial\mathcal{O}$

to $\mathcal{O}$ would make it solvable, which all seven sources refute. Hence there is no formal oscillator: $\partial y = iy$ needs a finite primitive of 1, and those are $\omega + c$, all infinite, so $\partial^2 y + y = 0$ factors as $(\partial - i)(\partial + i)y = 0$ and has only $y = 0$. Also: the purely infinite part of the accumulated phase as a complete gauge invariant in a four-term exact sequence; a sharp threshold over real-exponent Hahn fields, solvable iff $\min \text{supp } b > 1$, with the strictness at exponent 1 forced by the $\log \omega$ term and a counterexample pair showing the criterion cannot be globalized by valuation; Picard–Vessiot extensions with diagonal phase tori; and a coherent nonlinear initial-value theorem. *Merged; the hazard was a word.* Three different objects were all called “phase” — the finite angle in $\mathcal{O}$, the infinitesimal residue in $\mathfrak{m}$ left after removing a constant unit, and the accumulated phase, which is no angle at all and leaves $\mathcal{O}$ whenever its purely infinite part is nonzero. One source printed two of them twenty lines apart. The merged report fixes three symbols, relates them rather than identifying them, and prints the tempting false sentence beside the true one. The letter J , which meant three unrelated things across the batch, is retired.

17. Rank-one non-Archimedean analytic geometry inside the surcomplex numbers

[surcomplex/rank-one-berkovich/article.pdf](#)

from [surcomplex_rank_one_berkovich.zip](#)

The report devoted to intrinsic rank-one nonarchimedean analytic convergence. Inside the rank-one workspace $K = \mathbb{C}((t^{\mathbb{R}}))$ the absolute value $|a|_v = \exp(-v(a))$ is genuine and K is spherically complete, complete and algebraically closed, so it is a legitimate base for Tate algebras and Berkovich spaces. Proves strict inclusions among the Tate, polynomial-coefficient Hahn, common-domain, radius-free germ, formal-coefficient and open-disk analytic rings with named witnesses, extending the ring table of [surcomplex/analytic-geometry](#); gives a constructive one-variable preparation theorem, principal-ideal and finite-quotient results, analytic root counts and a valuation Rouché theorem; constructs the Berkovich disk points and the annulus retraction; and contrasts the contractible annulus spectrum with its nonzero analytic de Rham class. The worked case study is the entire series $F(Z) = \sum_{n \geq 0} t^{n^2} Z^n$, taken from [foundations-and-computation/foundations](#) and credited there: its full zero geometry is settled, and the first correction of the m th normalized root occurs at $t^{m(m+1)}$. *This does not contradict the no-convergence theorems of [surcomplex/analysis](#).* The convergence here is in the intrinsic valuation topology of one fixed rank-one field, not the fine topology induced from $\mathbf{No}[i]$; and at $Z = t^{-\omega}$ the exponents $n^2 - n\omega$ strictly decrease, so the support is not well ordered and F is not an all-surcomplex entire function — the obstruction agrees with all-scale rigidity. The original audit credits the Berkovich *comparison* in [surcomplex/contours-and-stokes](#); its development-gap claim concerns that inspected snapshot, not an exhaustive search of the current collection.

18. Surcomplex spectral theory: singular values, infinitesimal rank, multiscale stability

[surcomplex/spectral-theory/article.pdf](#)

from two manuscripts, delivered 21 September 2026

Finite-dimensional matrix algebra, singular scales and principal-root fields. Proves the finite spectral theorem and the singular value decomposition over a real closed field and its complexification *without compactness*: real closedness suffices, and the proofs exhibit the extremal witnesses rather than extracting them from a compactness argument, so Courant–Fischer and Ky Fan hold with extrema *attained*. On that base: positive square roots and polar decomposition, generalized Hermitian eigenproblems, Moore–Penrose inverses, least squares, best low-rank approximation, and conditioning. The scale-sensitive core identifies singular-value valuations with successive differences of minimum minor valuations, and recovers both those valuations and the leading singular coefficients from positive Gram coefficients one scale block at a time. Applications include an intrinsic rank filtration, root-free scale elimination, perturbation precision certificates, nonnormal pseudospectra, gap-sensitive spectral projectors and Schur-complement corrections. A second manuscript supplies Hermitian and normal spectral descent and the SVD over arbitrary set-sized ordered exponent groups, preserving the subgroup generated by the input supports

through a finite splitting tree. For a positive semidefinite matrix and integer $q \geq 2$, the principal q th root generates the field with exponent group $\Gamma + \sum_{k=1}^{\text{rank } A} \mathbb{Z}\Delta_k(A)/q$; the degree is the order of the subgroup generated by the determinantal minima $\Delta_k(A)$ modulo $q\Gamma$. Its trace generates the same extension. At $q = 2$, the condition $\Delta_k(A) \in 2\Gamma$ for $1 \leq k \leq \text{rank } A$ is also the exact Gram-factorization criterion. The targeted historical audit explains the report's organization; it does not establish absence of every related result elsewhere in the collection.

19. Surcomplex dynamics: linearization thresholds, periods, and normal forms

[surcomplex/dynamics-and-normal-forms/article.pdf](#)

from seven manuscripts, delivered 21 September 2026

Local and global dynamics over complex Hahn fields $K = \mathbb{C}((t^\Gamma))$ realized inside $\mathbf{No}[i]$, with Γ a set-sized ordered abelian group of *arbitrary* valuation rank. Seven manuscripts, and the report is their *union*: six are germ and fixed-point dynamics — multipliers, return iterates, periods, a Hamiltonian at the origin of a phase halo — and the seventh is the quasi-periodic half, a cohomological equation on a real torus whose frequency vector has *surreal* components and whose divisors $k \cdot \Upsilon$ are indexed by an integer lattice. The second half is genuinely additive: nothing in the first six carries a resonance lattice, a cohomological equation, a Fourier divisor or a conjugacy to a constant vector field, and the seventh carries no multiplier, no return iterate and no period.

Read Section 2 first. Five *inequivalent* conditions travelled under the single word *resonance* across these manuscripts and six under *small divisor*, and two groups of them used *radius* with **opposite** monotonicity; unifying the notation without separating the notions would have produced statements that read correctly and are false. Section 2 gives each variant its own name and symbol, every threshold statement names which variant and which coefficient category it holds in, and Section 2.5 is a theorem-by-theorem tag ledger. In particular lattice resonance is *not* frequency resonance — the latter is a condition on ordinary complex ω with divisors graded by Taylor degree and Hahn valuation zero, the former concerns surreal components with divisors graded by the Fourier index and of nonzero valuation — and the finitely many visible divisor valuations are *not* a spectrum in any sense this collection uses. The headline of the quasi-periodic half: a frequency vector with d surreal components has *at most* d arithmetically visible divisor valuations, identified by a canonical descending flag of saturated integer resonance lattices of strictly dropping rank, with one explicitly constructed well-ordered set containing the support of *every* reciprocal nonzero divisor; an exact stratified subexponential criterion, necessary and sufficient for universal solvability of the cohomological equation over a single fixed torus strip; an arbitrary-order lifting obstruction; and a unique normalized mean-zero conjugacy to a constant vector field proved by marked support words rather than by sequential valuation convergence. From the first half: the exact maximal centered valuation ball read off the finite q th return iterate rather than off f , the radius law $\text{rad} = R \exp(-(k+1)\sigma)$ with its trichotomy, a five-row coefficient category table with per-row necessity witnessed at a single exponent, an algebraicity dichotomy over $\mathbb{C}(w)((t^\Gamma))$ with its monodromy and cover obstruction, periods as a complete conjugacy invariant with an explicit moduli space, and a gauge-normalized symplectic normal form carrying no arithmetic bound at all.

20. Nonabelian support obstructions: a matrix Cousin criterion and the surcomplex Riemann–Hilbert problem

[surcomplex/nonabelian-support/article.pdf](#)

from `surcomplex_nonabelian_support` + `surcomplex_riemann_hilbert`

One coefficient category studied from two sides: matrices congruent to the identity modulo positive Hahn exponents, $I_r + \text{Mat}_r(J_\Gamma)$, over a set-sized and possibly non-Archimedean value group Γ , with ordinary holomorphic coefficient functions on a *single fixed* ordinary complex domain; when Γ is an ordered subgroup of the surreals, $t^\gamma = \omega^{-\gamma}$ gives the surcomplex reading. Part I solves the multiplicative gluing problem — a matrix Cousin problem on a puncture-and-disk cover of a connected open Riemann surface — and develops the resulting integral vector bundles and the Stein branch; Part II solves the inverse monodromy problem for near-identity representations

of the fundamental group of a plane with a closed discrete puncture set. **The two headline theorems both read “the union of the supports is well ordered” and are *not* the same criterion**, and a whole section exists to keep them apart: the gluing criterion is about the supports of the *normalized polar factors* $\text{Pol}(G_a) - I_r$, each transition matrix being factored independently and first. Well-ordering of the full raw transition support is sufficient but not necessary; raw singular-coefficient support fails in *both* directions. The realizability criterion is about the *raw* monodromy matrices $\rho(\ell_a) - I_r$, where the raw support is the tested support, whose well-ordering is basis- and basepoint-independent. The trap is printed where a reader would otherwise generalize: a rank-three family over $\Gamma = \mathbb{Q}$ with determinant one, a single raw singular exponent and a *splitting* abelianized cocycle admits no splitting, because the polar factors are forced to carry a descending set; adding one term introduces that descending set among the raw poles and thereby *removes* the obstruction; and the true matrix logarithm is not a repair, its coefficient being off by a factor of $1/2$. Also the rank-one recovery of this collection’s scalar Cousin criterion, the right-torsor description of all splittings, absoluteness under ordered value-group extension, and a depth hierarchy giving for every $r \geq 3$ a cocycle trivial modulo the last central subgroup and nonsplit in full positive rank. Every \label in its source carries the prefix nab:.

21. Cofinality, factorization and scalar extension for entire Hahn functions at arbitrary rank

[surcomplex/entire-functions-at-arbitrary-rank/article.pdf](#)

from two manuscripts, delivered 21 September 2026

Ordinary power series $\sum a_n Z^n$ over a *fixed* Hahn field $K_\Gamma = \mathbb{C}((t^\Gamma))$ with Γ nonzero, set-sized, divisible and of arbitrary valuation rank, where *entire* means strongly Hahn-summable at every point of that one field; both sources explicitly disclaim all of **No**[*i*]. Two manuscripts that were the same paper twice on the engine and complementary on the payload — the same exact coefficient criterion, cofinality dichotomy, support-controlled preparation theorem, arbitrary-rank root counts, genus-zero canonical product, factorization by zeros and sharp change-of-workspace criterion, by the same witness — so each is printed once and both payloads are kept in full. **The trichotomy.** The entire ring E_Γ is a GCD domain with units exactly $K_\Gamma^\times$, and exactly one of: uncountable cofinality, where only polynomials are entire and the ring is a PID; an *order unit* exists, where nonpolynomial series exist and the ring is Bézout, non-Noetherian, with full Hermite interpolation; or countable cofinality with no order unit, where the ring is GCD but **not** Bézout — two canonical products with disjoint simple zero sets and gcd 1 generating a proper ideal, the obstruction an exact valuation computation against a growth barrier every entire function obeys. The middle case is *sharp*, not merely sufficient: a nonconstant valuation-restricted series is invertible iff its least coefficient gap is a positive order unit, necessity proved by coarsening. For $\Gamma = 0$ the strong-summation convention gives $\mathbb{C}[Z]$. **The scalar-extension package.** For an extension of divisible ordered groups $\Gamma \subseteq \Delta$, *every* nonpolynomial entire series over K_Γ has the *same* exact strong evaluation domain in K_Δ — the valuation ring of v_Δ coarsened by the convex hull of Γ , whose residue field is a Hahn field rather than $\mathbb{C}$ — entireness survives exactly for cofinal extensions, no new zeros appear in the surviving domain, and the boundary is intrinsic. The two sources’ extension statements look like disagreement about whether the function survives and are the coarse and the exact form of one fact: “not entire over K_Δ ” means precisely that the domain is a proper subring, and the extension theorem computes which. One name is kept for the order invariant, with the synonym recorded once; it is not the same invariant as cofinality, and that gap is exactly the third case. Extends the ring table of [surcomplex/rank-one-berkovich](#), which lands in the Bézout case so the third case is invisible there; its whole-class rigidity corollary is presented as a *re-derivation* from the extension theorem, crediting [surcomplex/analysis](#) for all-scale polynomial rigidity and [foundations-and-computation/foundations](#) for the sign-flipped pair over $\mathbb{C}((t^\mathbb{Q}))$ and for the localization principle it consumes. Rank is not the driver: a rank-two group with an order unit is Bézout, and the direct sum of copies of $\mathbb{Q}$ indexed by ω_1 admits only polynomials.

3 Surquaternions

One report, which is where commutativity stops. It is filed on its own because commutative arguments from `docs/surcomplex` require review on moving to a division algebra: conjugation, multiplicativity of the modulus, valuation and leading coefficients, polynomial evaluation, matrix adjoints, exponential laws and every chain rule need their scalar and slice hypotheses checked. The report proves the additional noncommutative assertions with factor order retained.

22. Surquaternions: algebra, infinitesimal geometry, and support-controlled analysis

`surquaternions/surquaternions/article.pdf`

`from surquaternions_article.zip + surquaternions_article(1).zip`

Hamilton’s quaternion division algebra over the surreals, $\mathbb{H}_{\mathbf{No}} = \mathbf{No} + \mathbf{No}i + \mathbf{No}j + \mathbf{No}k$, with central surreal coefficients, in three levels that are never silently mixed. *Finite algebra over an arbitrary real closed field*: division, conjugation, the field-valued modulus, complex slices and centralizers, conjugacy spheres, explicit square roots, the spin/rotation description with two independent surjectivity proofs, Cayley and stereographic presentations of one rational chart, quaternionic Möbius maps, one-sided polynomials with a constructive fundamental theorem and a complete zero-class algorithm, and finite-dimensional Hermitian spectral theory with SVD and polar decomposition. *Set-sized Hahn workspaces* $D_{\Gamma} = \mathbb{H}(\mathbb{R}((t^{\Gamma})))$: normal forms with a **noncommutative** residue algebra, exact multiplicative valuation and leading coefficients, the all-lengths Neumann support lemma proved through Higman’s finite-word lemma, the exact geometric inverse with its truncation certificate, the infinitesimal Hahn exponential and logarithm, BCH at positive valuation, the filtered nonabelian rotation group with its graded bracket, a spectral-projector estimate at arbitrary surreal gap scale with a sharp two-scale example, support-controlled implicit lifting, and an exact square-root conditioning analysis. *The full surreal class*: finite angles at arbitrary radius, the Ehrlich–Kaplan global phase normalization, the global radial exponential with its complete logarithm fibres and four-coordinate differential — rank-two critical spheres at positive radii in $\pi\mathbf{Oz}$, $\mathbf{Oz}$ the omnific integers, the infinite radius ω among them — the componentwise Berarducci–Mantova derivation with a no-oscillation theorem and its chain-rule corollary, and coefficientwise slice, Cauchy and Fueter analysis on explicitly named coefficient spaces. **Section 2 is the ledger and should be read first**: three exponentials and four derivative-like operators carry overlapping names here, each satisfying a *different* law on a *different* domain, and the section fixes one name and one symbol per object and tabulates which satisfies which. It also explains why this report’s statement that the radial exponential is *not* an additive-to-multiplicative homomorphism is consistent with the physics report’s correct claim that the *commutative* Ehrlich–Kaplan exponential *is* one: the restriction to any single slice is exactly that commutative exponential and the law holds there, and it can fail across *noncommuting* arguments, where no common slice exists — witness $\text{Exp}(\omega i + \pi j) = 1 + (\pi^2/2\omega)i + O(\omega^{-2})$ against $\text{Exp}(\omega i)\text{Exp}(\pi j) = -1$. A merged sentence blurring the two would manufacture a false contradiction across the collection. *Merged; roughly two thirds of the mathematics was common to both sources and is printed once*. Every `\label` in its source carries the prefix `squat:`. It makes no claim about any empirical subject.

4 Physics

One report, and the only one in the collection whose subject is not mathematics. It is filed on its own because its entire value is a firewall between what is proved and what is proposed, and because its conclusion is negative.

23. Surreal scalars in theoretical physics: black-hole singularities, asymptotic structure, and the limits of replacing the scalar field

[physics/surreal-scalars-and-spacetime/article.pdf](#)

from three manuscripts, delivered 21 September 2026

The sentence this report exists to refuse, in any form, is that surreal numbers resolve the black-hole singularity. Surreal and surcomplex scalars make the hierarchy of quantities near a gravitational singularity exactly representable; they do *not* make the spacetime regular. Four facts, each re-derived from the metric for the merge rather than copied from a source: the Kretschmann scalar of Schwarzschild is exactly $K = 48M^2/r^6$; under a formal radial change $r = s^q w(s)$, with integer $q \geq 1$, $w(0) \neq 0$ and $M \neq 0$, and $w \in \mathbb{R}[[s]]$, the pole order becomes exactly $6q$ with leading coefficient $48M^2 w(0)^{-6}$. The recorded checks use $w(s) = 1 + s$, whose constant term is one; this normalized case preserves $48M^2$. No substitution of the stated form removes the pole; the proper time from rest at R to $r = 0$ is $(\pi/2)\sqrt{R^3/2M}$, which for *infinitesimal* R is infinitesimal and hence still finite, so the infall terminates either way; and the tidal component $2M(2M - r)/r^4$ is unlimited at infinitesimal r . The exact identity $r^6 K = 48m^2$ has *no* field-valued solution at $r = 0$ in any characteristic-zero field, not merely in $\mathbb{R}$: evaluating at a nonzero infinitesimal radius names another point of a punctured domain and supplies no missing endpoint, no continuation law and no observable rule. What is true, and is what the report is for, is that the surreal/Hahn language records the power *and* the coefficient of every divergence exactly, keeps competing scales separate, and detects when an expansion has been used outside its regime — a gain in representation and nothing more. **The claim-status firewall** is the rest of its value: every theorem header and every substantive paragraph carries one of four visible tags — exact symbolic identity, checked exactly and never to a tolerance, which is the entire content of what the programs establish; conditional theorem, with hypotheses stated in full and *established for no physical model*, nothing in the tier machine-checked; assessment, meaning an argument and not a result; and imported, a provenance marker rather than a fourth tier. The article’s own tag counts are 53, 51, 140 and 34. *Merged; duplication between the three sources ran to about two thirds, all three independently rebuilding the static-spherical connection and the full Riemann tensor and re-deriving the same Schwarzschild quantities from the metric alone.* No priority claim; not refereed.

5 Foundations and computation

Three reports that take the rest of the collection as their object. None belongs under **surreal** or **surcomplex**, because all three range over the whole subject. Each is a merge of three manuscripts delivered together, the first two on 21 September 2026, and in those two the hazard was not overlap but a collision of near-identical names.

24. Foundations for surreal and surcomplex mathematics

[foundations-and-computation/foundations/article.pdf](#)

from surreal_surcomplex_foundations.zip + surcomplex_foundations.zip + surreal_foundations.zip

In what foundation does this mathematics legally live, and how would a proof assistant encode it? Compares ZF/ZFC with definable classes, GB/GBC/NBG, Kelley–Morse, Grothendieck universes and TG, Feferman reflection, constructive set theory, dependent type theory, univalent foundations and native Lean, and proves the size obstructions each must respect: no unrestricted cut operation can accept every subclass of its own carrier, the surreals are not Dedekind complete, and a universe cannot cut itself. The positive counterpart is a localization theorem — every *set* of surcomplex numbers already lies in one divisible set-sized workspace $K_\Gamma = \mathbb{C}((t^\Gamma))$ — with a finite-algebra descent theorem explaining why a finite zero scheme gains no points in a larger workspace. Carries two series over $\mathbb{C}((t^\mathbb{Q}))$ that differ only in the sign of an exponent and reach opposite conclusions: $\sum t^{-n^2} X^n$ is summable at no nonzero argument, while $\sum t^{+n^2} z^n$ is summable everywhere and is a *nonpolynomial* function coherent at every radius — which does not refute the all-scale rigidity

theorem of `surcomplex/analysis`, stated for class functions on the whole `No[i]`, but shows it does not transfer to a fixed value group. *Merged; the danger was aggregation. All three manuscripts read the same Lean repository at the same revision on the same day, so their agreement is one act of source reading performed three times, not three confirmations. The findings are printed once and the verification boundary has its own early section.* The original merge compiled three illustrative Lean files and inspected their axiom dependencies. Those historical checks are distinct from the later development mapped in `FORMALIZATION.md`. The maintained article also makes the universe-relative localization hypotheses explicit, expands the positive-support recursion proof, and separates intrinsic convergence from full fine topology.

25. Surreal and surcomplex numbers in symbolic computer algebra

[foundations-and-computation/computer-algebra/article.pdf](#)

from `surreal_surcomplex_cas.zip` and two same-named archives

What can actually be computed exactly, and what only approximated or denoted? Separates finite exact denotation, effective coefficient access, decidable equality and order, and certified approximation into capability tiers, then works through an ordered rational-monomial core, real-algebraic closure, certified Hahn series, formal germs, analytic lifts, transseries, Newton profiles, Hensel lifting, finite quotient algebras, residues, and a proposed Wolfram Language architecture. *Merged; the danger was three prototypes with one name.* Each manuscript ships an executable Wolfram package, two of them near-anagrams of each other, two exporting six identical public symbol names with incompatible signatures, and all three calling themselves the rational monomial core. They are three different objects: coefficients $\mathbb{Q}$ in one and $\mathbb{Q}(i)$ in the others; exponent lattices $\mathbb{Z}^d$ against $\mathbb{Q}^r$, so a ramified root is representable in one and not the next; and two of them order their monomial variables by opposite conventions, requiring the corresponding translation of monomial-order inequalities. The report never writes “the prototype” and tabulates the divergences instead. All three check suites were rerun for this repository and reproduce exactly — 34/34, 110/110 and 88/88 in Wolfram, 69/69 and 323/323 in Python — but one package cannot be loaded at all as shipped: an unclosed bracket stops it parsing, which is why its own record submits definitions rather than loading the file.

26. Computable surreal numbers, merged and reconciled

[foundations-and-computation/computable-surreals/article.pdf](#)

from `computable_surreal_numbers.zip` + two same-named archives

Which surreal numbers admit an algorithm, and in which representation? Three manuscripts arrived with two different numerical representations of the same objects — one using effective rational left-finite series with finite candidate covers, two using computable bounded-denominator Puiseux series — and the report is one integrated development rather than a concatenation: it preserves the distinct representations instead of picking one, supplies an explicit reconciliation ledger disposing of every divergence, keeps complementary proofs where the two routes prove the same statement by different means, and separates mathematical existence from the existence of a uniform algorithm throughout. *Merged; and this is the one report whose source manuscripts were never redistributed with it in the first place.* The ledger cites them by their original $\text{\TeX}$ line numbers and `data/provenance-manifest.json` records their archive identities, file hashes and line counts, so every disposition is traceable against a copy obtained elsewhere. Four verification suites are rerun and pass — 4,967 exact comparisons, 18 named unit tests, 266 exact checks, and 4,366 exact cross-model and cover-contract checks in the updated reconciliation suite. The latter compare dense common-grid Puiseux convolution against sparse rational convolution, candidate covers with extra zeros and nonnested lists (including negative zero candidates in the geometric-series construction), the finite-jet root transformation, derivative and primitive identities, and finite instances of the two unbounded-denominator support examples. *The four units mean different things and are deliberately not added together.* A finite test proves no infinite theorem, no real closedness, no computability classification and no undecidability reduction; the synthetic halting fixtures in the original programs do not compute the halting set; the package implements neither

arbitrary computable reals nor arbitrary infinite surreal arithmetic; and there is no Lean-checked formalization and no literature-wide priority claim.